# Solution to Exercise 02 — Analysis on Manifolds, 80416

April 21, 2026

# Question 1

We will draw the following paths and calculate the line or path integrals for each following path and vector fields $F : \mathbb{R}^n \to \mathbb{R}^n$ or functions $f : \mathbb{R}^n \to \mathbb{R}$.
*Reminder:* The line integral of $F$ along $\gamma$ is defined by

$$\int_\gamma F \, \mathrm{d}l = \int_0^1 \langle F(\gamma(t)), \dot{\gamma}(t) \rangle \, \mathrm{d}t$$

The path integral of $f$ along $\gamma$ with respect to the length element is defined by

$$\int_\gamma f \, \mathrm{d}s = \int_a^b f(\gamma(t)) \|\dot{\gamma}(t)\| \, \mathrm{d}t$$

So in each part we calculate each component in these integrals.

## Subquestion 1

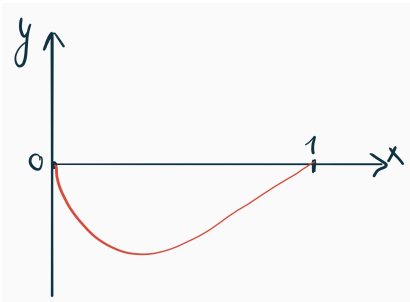
$\gamma : [0,1] \to \mathbb{R}^2,\ \gamma(t) = (t^2, t^2 - t),\ F(x,y) = (x^2 y, y - 3x)$ and we will calculate $\int_\gamma F \, \mathrm{d}l$.
*Solution*:

$$\dot{\gamma}(t) = (2t, 2t - 1)$$
$$F(\gamma(t)) = F(t^2, t^2 - 2) = \left( (t^2)^2 (t^2 - t), t^2 - t - 3t^2 \right) = (t^6 - t^5, -2t^2 - t)$$

So the full integrand is

$$\int_\gamma F \, \mathrm{d}l = \int_0^1 (t^6 - t^5, -2t^2 - t) \cdot (2t, 2t - 1) \, \mathrm{d}t = \int_0^1 2t^7 - 2t^6 - 4t^3 + t \, \mathrm{d}t = \left[ \frac{t^8}{4} - \frac{2t^7}{7} - t^4 + \frac{t^2}{2} \right]_{t=0}^{t=1} = -\frac{15}{28}$$

For the drawing: we have a 2D curve that starts at the origin (0,0) and ends at (1,0) and from $\dot{\gamma}(0) = (0, -1)$ we get that the curve starts by moving straight down from the origin.
Furthermore, from $\dot{\gamma}(t)$ we can see that the curve has a horizontal tangent at $t = \frac{1}{2}$ and this is the lowest point on the curve $\left(\frac{1}{4}, -\frac{1}{4}\right)$ and at the end we get that $\dot{\gamma}(1) = (2, 1)$ so the curve finishes while moving up and to the right.
From $x = t^2$ we know that $t = \sqrt{x}$ since $t \in [0,1]$ so the $y$ equation becomes $y = x - \sqrt{x}$ so the drawing is




$\square$

## Subquestion 2

$\gamma : [0,1] \to \mathbb{R}^3,\ \gamma(t) = (t, \cos(t), \sin(t)),\ F(x,y,z) = (0, -z, y)$ and we will calculate $\int_\gamma F \, \mathrm{d}l$.

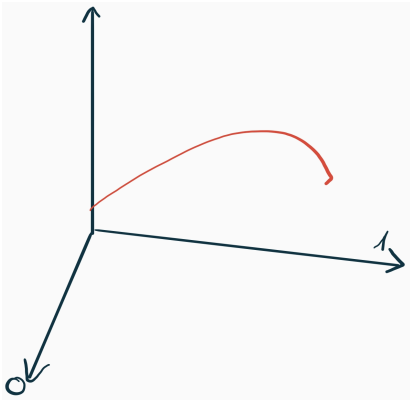
*Solution*:

$$\dot{\gamma}(t) = (1, -\sin(t), \cos(t))$$
$$F(\gamma(t)) = F(t, \cos(t), \sin(t)) = (0, -\sin(t), \cos(t))$$

So the full integrand is

$$\int_\gamma F \, \mathrm{d}l = \int_0^1 (0, -\sin(t), \cos(t)) \cdot (1, -\sin(t), \cos(t)) \, \mathrm{d}t = \int_0^1 0 + \sin^2(t) + \sin^2(t) \, \mathrm{d}t = \int_0^1 1 \, \mathrm{d}t = 1$$

For the drawing: This is a 3D curve and we notice that $y^2 + z^2 = \cos^2(t) + \sin^2(t) = 1$ so in the $yz$-plane the path is the unit circle and since the $x$-coordinate is exactly $t$, the path moves forward along the $x$-axis as it "circles" the $yz$-plane. Furthermore, since $\|\dot{\gamma}(t)\| = \sqrt{2}$ we understand that this is a smooth $C^1$ curve so that drawing is




$\square$

## Subquestion 3

$\gamma : \left[\frac{1}{4}, \frac{3}{4}\right] \to \mathbb{R}^2,\ \gamma(t) = \left(2t-1, \sqrt{1-(2t-1)^2}\right),\ f(x,y) = xy^4$ and we will calculate $\int_\gamma f \,\mathrm{d}s$.

*Solution*: First we do variable change for $x = 2t-1$ so $\gamma(t) = \left(x, \sqrt{1-x^2}\right)$ so this is the upper semicircle of radius 1 with start point at $t = \frac{1}{4}$ which means $x = -\frac{1}{2}, \gamma\left(\frac{1}{4}\right) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ and end point at $t = \frac{3}{4}$ which means $x = \frac{1}{2}, \gamma\left(\frac{3}{4}\right) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. We will use the hint and use the arc-length reparameterization and work as we saw in the recitation:

$$\dot{\gamma}(t) = \left(2, \frac{-2(2t-1)}{\sqrt{1-(2t-1)^2}}\right)$$

$$\|\dot{\gamma}(t)\| = \sqrt{2^2 + \left(\frac{-2(2t-1)}{\sqrt{1-(2t-1)^2}}\right)^2} = \frac{2}{\sqrt{1-(2t-1)^2}}$$

$$l(\gamma) = \int_{\frac{1}{4}}^{\frac{3}{4}} \|\dot{\gamma}(t)\| \,\mathrm{d}t = \int_{\frac{1}{4}}^{\frac{3}{4}} \frac{2}{1-(2t-1)^2} \,\mathrm{d}t \underset{\substack{u=2t-1 \\ \mathrm{d}u=2\,\mathrm{d}t}}{=} \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{1-u^2}} \,\mathrm{d}u = [\arcsin(u)]_{u=-\frac{1}{2}}^{u=\frac{1}{2}} \frac{\pi}{6} - \left(-\frac{\pi}{6}\right) = \frac{\pi}{3}$$

We define the function $\varphi : \left[\frac{1}{4}, \frac{3}{4}\right] \to \left[0, \frac{\pi}{3}\right]$ by

$$\varphi(t) = \int_{-\frac{1}{4}}^{t} \|\dot{\gamma}(s)\| \,\mathrm{d}s = \int_{\frac{1}{4}}^{t} \frac{2}{\sqrt{1-(2s-1)^2}} \,\mathrm{d}s = [\arcsin(2s-1)]_{s=\frac{1}{4}}^{s=t} = \arcsin(2t-1) - \arcsin\left(-\frac{1}{2}\right) = \arcsin(2t-1) + \frac{\pi}{6}$$

Now we need to find $\varphi^{-1}$ so we solve for $t$ in terms of arc length $s \in \left[0, \frac{\pi}{3}\right]$

$$s = \arcsin(2t-1) + \frac{\pi}{6} \iff s - \frac{\pi}{6} = \arcsin(2t-1) \iff 2t-1 = \sin\left(s - \frac{\pi}{6}\right) \implies t = \varphi^{-1}(s) = \frac{1}{2}\left(\sin\left(s-\frac{\pi}{6}\right) + 1\right)$$

We remember that $y \geq 0$ on our interval and we gets that

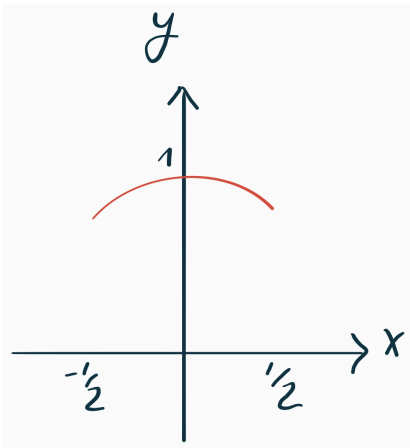
$$x(s) = 2\varphi^{-1}(s) - 1 = \sin\left(s - \frac{\pi}{6}\right) \qquad y(s) = \sqrt{1-\sin^2\left(s-\frac{\pi}{6}\right)} = \cos\left(s-\frac{\pi}{6}\right)$$

$$\implies \overline{\gamma} = \left(\sin\left(s-\frac{\pi}{6}\right), \cos\left(s-\frac{\pi}{6}\right)\right)$$

We saw that $\int_\gamma f \,\mathrm{d}s$ is invariant under re-parametrizations so

$$\int_\gamma f \,\mathrm{d}s = \int_0^{\frac{\pi}{3}} f(\overline{\gamma}(s)) \|\dot{\overline{\gamma}}(s)\| \,\mathrm{d}s = \int_0^{\frac{\pi}{3}} \sin\left(s-\frac{\pi}{6}\right)\cos^4\left(s-\frac{\pi}{6}\right) \mathrm{d}s \underset{\substack{v=\cos\left(s-\frac{\pi}{6}\right) \\ \mathrm{d}v=-\sin\left(s-\frac{\pi}{6}\right)\mathrm{d}s}}{=} \int_{\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} -v^4 \,\mathrm{d}v = 0$$

For the drawing: since we already know that this is an arc of the upper unit circle from $x = -\frac{1}{2}$ to $x = \frac{1}{2}$ the drawing is




$\square$

## Question 2

Let $\gamma : [a, b] \to \mathbb{R}^n$ be a regular curve, and let $\overline{\gamma}$ be its arc length parameterization.
We will show that $l\left(\overline{\gamma} \restriction_{[s_1, s_2]}\right) = s_2 - s_1$ for $0 \leq s_1 \leq s_2 \leq l(\gamma)$ (meaning that the amount of length covered by $\overline{\gamma}$ within any interval is the length of the interval).
*Proof*: We defined the arc length parametrization of $\gamma$ as $\overline{\gamma} = \gamma \circ \varphi^{-1}$ where $\varphi : [a, b] \to [0, l(\gamma)]$ given by $\gamma(t) = \int_a^t \|\dot{\gamma}(s)\| \, \mathrm{d}s$ and we saw in the recitation a lemma that implies $\|\dot{\overline{\gamma}}\| \equiv 1$ hence

$$l\left(\overline{\gamma} \restriction_{[s_1, s_2]}\right) = \int_{s_1}^{s_2} \|\dot{\overline{\gamma}}(s)\| \, \mathrm{d}s = \int_{s_1}^{s_2} 1 \, \mathrm{d}s = s_2 - s_1$$

Where the last equality is from the definition of indicator function/The Fundamental Theorem of Calculus. $\square$

## Question 3

In this question we will prove the following theorem:
Let $\gamma : [a,b] \to \mathbb{R}^n$ be a regular $C^1$ curve such that $\gamma(a) = A, \gamma(b) = B$. Then $l(\gamma) \geq \|B - A\|$ and there is an equality if and only if $\forall t \in [a,b],\ \gamma(t) \in \{(1-s)A + sB \mid s \in [0,1]\}$.

### Subquestion 1

We will show that $l(\gamma) \geq \|B - A\|$.
*Proof*: Since $\gamma$ is differentiable, from The Fundamental Theorem of Calculus and the definition of $l(\gamma)$ we gets

$$B - A = \gamma(b) - \gamma(a) = \int_a^b \dot{\gamma}(t)\,\mathrm{d}t$$

By the triangle inequality for integrals

$$\|B - A\| = \left\| \int_a^b \dot{\gamma}(t)\,\mathrm{d}t \right\| \leq \int_a^b \|\dot{\gamma}(t)\|\,\mathrm{d}t = l(\gamma)$$

□

### Subquestion 2

Let $x, y \in \mathbb{R}^n$ and we will show that if $\|x + y\| = \|x\| + \|y\|$ then there exists $\alpha \geq 0$ such that $x = \alpha y$.
*Proof*: We start by squaring both sides and using the definition of inner product

$$\|x + y\|^2 = (\|x\| + \|y\|)^2 \iff \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 = \|x\|^2 + 2\|x\|\|y\| + \|y\|^2$$

Subtracting $\|x\|^2 + \|y\|^2$ from both sides and diving by 2 yields

$$\langle x, y\rangle = \|x\|\|y\|$$

From Cauchy-Schwarz inequality we gets $|\langle x, y\rangle| \leq \|x\|\|y\|$ and equality holds if and only if $x$ and $y$ are linearly dependent.
We found that $\langle x, y\rangle = \|x\|\|y\|$ so it means that $x$ and $y$ are linearly dependent and the angle between them is $0°$ which means that they are in the same direction.
Thus, there exists a scalar $\alpha$ such that $x = \alpha y$ and since $\langle x, y\rangle = \|x\|\|y\|$ is not negative then $\alpha \geq 0$. □

### Subquestion 3

We will show that for every $t_0 \in [a,b],\ l(\gamma) \geq \|\gamma(t_0) - A\| + \|B - \gamma(t_0)\|$.
*Proof*: let $t_0 \in [a,b]$, from the linearity of integrals

$$l(\gamma) = \int_a^b \|\dot{\gamma}(t)\|\,\mathrm{d}t = \int_a^{t_0} \|\dot{\gamma}(t)\|\,\mathrm{d}t + \int_{t_0}^b \|\dot{\gamma}(t)\|\,\mathrm{d}t$$

Part (1) implies that

$$l\left(\gamma \restriction_{[a,t_0]}\right) \geq \|\gamma(t_0) - \gamma(a)\| = \|\gamma(t_0) - A\|$$
$$l\left(\gamma \restriction_{[t_0,b]}\right) \geq \|\gamma(b) - \gamma(t_0)\| = \|B - \gamma(t_0)\|$$

We sum both inequalities

$$l(\gamma) = l\left(\gamma \restriction_{[a,t_0]}\right) + l\left(\gamma \restriction_{[t_0,b]}\right) \geq \|\gamma(t_0) - A\| + \|B - \gamma(t_0)\|$$

□

## Subquestion 4

We will use parts (2) and (3) to conclude that $l(\gamma) = \|B - A\|$ implies $\forall t \in [a, b],\ \gamma(t) \in \{(1-s)A + sB \mid s \in [0,1]\}$.
*Proof*: Assume that $l(\gamma) = \|B - A\|$ and from part (3) for any $t \in [a, b]$

$$\|B - A\| \geq \|\gamma(t) - A\| + \|B - \gamma(t)\|$$

From the triangle inequality

$$\|B - A\| \leq \|\gamma(t) - A\| + \|B - \gamma(t)\|$$

Thus

$$\|B - A\| = \|\gamma(t) - A\| + \|B - \gamma(t)\|$$

From part (1) we get $\alpha \geq 0$ so that

$$B - A = (\gamma(t) - A) + (B - \gamma(t)) \iff \gamma(t) - A = \alpha(B - \gamma(t))$$

We rearrange

$$\gamma(t) = \frac{A}{1+\alpha} + \frac{\alpha B}{1+\alpha}$$

Let $s = \frac{\alpha}{1+\alpha} \in [0, 1]$ so

$$\gamma(t) = (1-s)A + sB$$

$\square$

## Subquestion 5

We will prove that if $\forall t \in [a, b],\ \gamma(t) \in \{(1-s)A + sB \mid s \in [0,1]\}$ then $l(\gamma) = \|B - A\|$.
*Proof*: By the assumption, every point of the curves lies on the path from $A$ to $B$ so there exists a function $s(t) \in [0, 1]$ such that

$$\gamma(t) = (1 - s(t))A + s(t)B$$
$$\dot{\gamma}(t) = \dot{s}(t)(B - A)$$

So the length is given by

$$l(\gamma) = \int_a^b \|\dot{\gamma}(t)\| \,\mathrm{d}t = \int_a^b |\dot{s}(t)| \|B - A\| \,\mathrm{d}t = \|B - A\| \int_a^b |\dot{s}(t)| \,\mathrm{d}t$$

We notice that $s(t)$ is non-decreasing (since by definition, $\gamma(a) = A,\ \gamma(b) = B$ so $s(a) = 0,\ s(b) = 1$ and the curves moves along the path without reversing direction) and hence $|\dot{s}(t)| = \dot{s}(t)$ thus using The Fundamental Theorem of Calculus

$$l(\gamma) = \|B - A\| \int_a^b |\dot{s}(t)| \,\mathrm{d}t = \|B - A\| \int_a^b \dot{s}(t) \,\mathrm{d}t = \|B - A\|(s(b) - s(a)) = \|B - A\| \cdot 1 = \|B - A\|$$

$\square$

## Question 4

We call a vector field $F : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}^n$ a radial field if there exists a $C^1$ function $g : \mathbb{R}_+ \to \mathbb{R}$ such that

$$\forall x \in \mathbb{R}^n,\ F(x) = g(\|x\|)x$$

We will show that if $F : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}^n$ is a radial field then there exists a $f \in C^1,\ f : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}$ such that $F = \nabla f$.
*Proof*: Same as in the recitation, we notice that $F(x)$ depends only on $\|x\|$ and points in the direction $x$.
We look for a function $f$ that depends only on the distance to the origin so we define $f : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}$ as

$$f(x) = \int_0^{\|x\|} u \cdot g(u)\,\mathrm{d}u$$

We can integrate since $g \in C^1$ so $u \cdot g(u) \in C^1$ From the Chain Rule we have

$$\frac{\mathrm{d}f}{\mathrm{d}x_i} = (\|x\| \cdot g(\|x\|)) \cdot \frac{x_i}{\|x\|} = g(\|x\|)x_i$$

Therefore

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, ..., \frac{\partial f}{\partial x_n}\right) = (g(\|x\|)x_1, ..., g(\|x\|)x_n) = g(\|x\|) \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = g(\|x\|)x = F(x)$$

$\square$

## Question 5

Let $\Omega \subseteq \mathbb{R}^n$ be an open set in $\mathbb{R}^n$. We call a smooth function $\varphi : [0,T] \times \Omega \to \Omega$ a flow if:

1. For all $t \in [0,T]$ the map $\varphi_t : \Omega \to \Omega$ defined by $\varphi_t(x) = \varphi(t,x)$ is a diffeomorphism
2. For all $x \in \Omega,\ \varphi(0,x) = x$

Let $F : [0,T] \times \Omega \to \mathbb{R}^n$ be a smooth function (this function induces a vector field on $\Omega$ for every $t \in [0,T]$).
We say that $F$ is compatible with the flow $\varphi$ if:

$$\forall x \in \mathbb{R}^n,\ \frac{\mathrm{d}}{\mathrm{d}t}\varphi(t,x) = F(t,\varphi(t,x))$$

And assume that $\forall s,t \in [0,T]$ and $\forall x \in \Omega,\ \varphi(s+t,x) = \varphi(t,\varphi(s,x))$ and that $F$ is compatible with $\varphi$.
We will show that $\forall t,s \in [0,T]$ and $\forall x \in \Omega,\ F(t,x) = F(s,x)$.
*Proof*: In other words, we want to show that $F$ does not depend on time.
Fix $s,t,x$, from our assumption

$$\varphi(s+t,x) = \varphi(t,\varphi(s,x))$$

We differentiate with respect to $t$ each side:
Since $F$ compatible with $\varphi$

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi(s+t,x) = F(s+t,\varphi(s+t,x))$$

On the right side since $\varphi(s,x)$ is a constant

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi(t,\varphi(s,x)) = F(t,\varphi(t,\varphi(s,x))) = F(t,\varphi(s+t,x))$$

Where the last equality is from out assumption.
So we have

$$F(s+t,\varphi(s+t,x)) = F(t,\varphi(s+t,x))$$

Since $\varphi_{s+t}$ is a diffeomorphism, every $y \in \Omega$ can be written as

$$y = \varphi(s+t,x)$$

So our equation above becomes for all $y \in \Omega$

$$(\star)\ F(s+t,y) = F(t,y)$$

Now we fix $x$ and want aim to show that $F(t,x) = F(s,x)$.
From $(\star)$ using $y = x, t = 0$ we get

$$F(s+0,x) = F(s,x) = F(0,x)$$

Similarly, we get that $F(t,x) = F(0,x)$ so $F(s,x) = F(t,x)$ for all $s,t,x$ and we are done. $\square$